Problem Set 1– Advanced Macroeconomics

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Exercise 1 – Business Cycle Data

United States

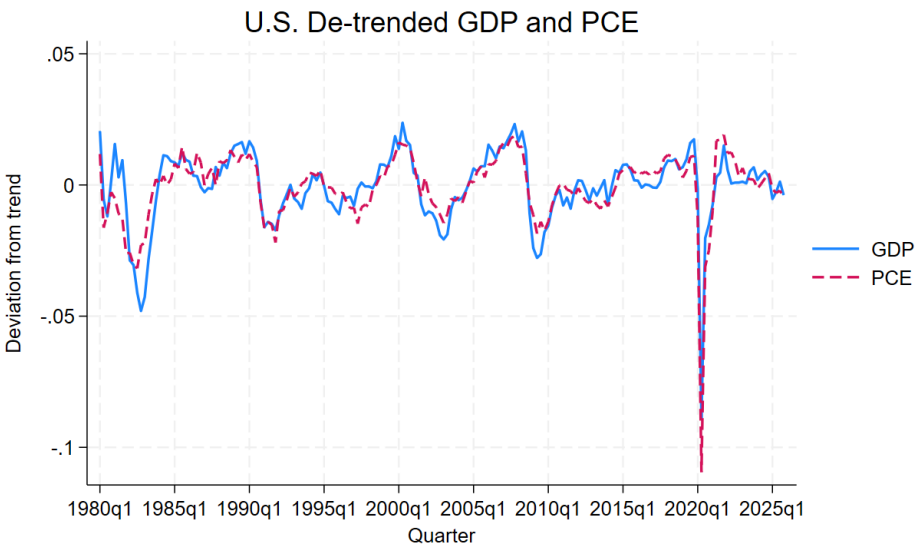

Figure 1: De-trended GDP and consumption in the United States (HP filter, $\lambda = 1600$).

Table 1: Covariance matrix (United States)

	us_pce_cycle	us_gdp_cycle
us_pce_cycle	0.000171	
us_gdp_cycle	0.000157	0.000189

Table 2: Correlation matrix (United States)

	us_pce_cycle	us_gdp_cycle
us_pce_cycle	1.0000	
us_gdp_cycle	0.8718	1.0000

Comment. Figure 1 shows that the de-trended series for GDP and consumption move closely together over time. Their co-movement is confirmed by the correlation coefficient reported in Table 2, which is equal to 0.8718. This indicates a strong positive relationship between the two cyclical components. Moreover, aside from the sharp contraction associated with the Covid-19 shock, both series display relatively limited volatility, which is consistent with the small variances and covariance reported in Table 1.

France

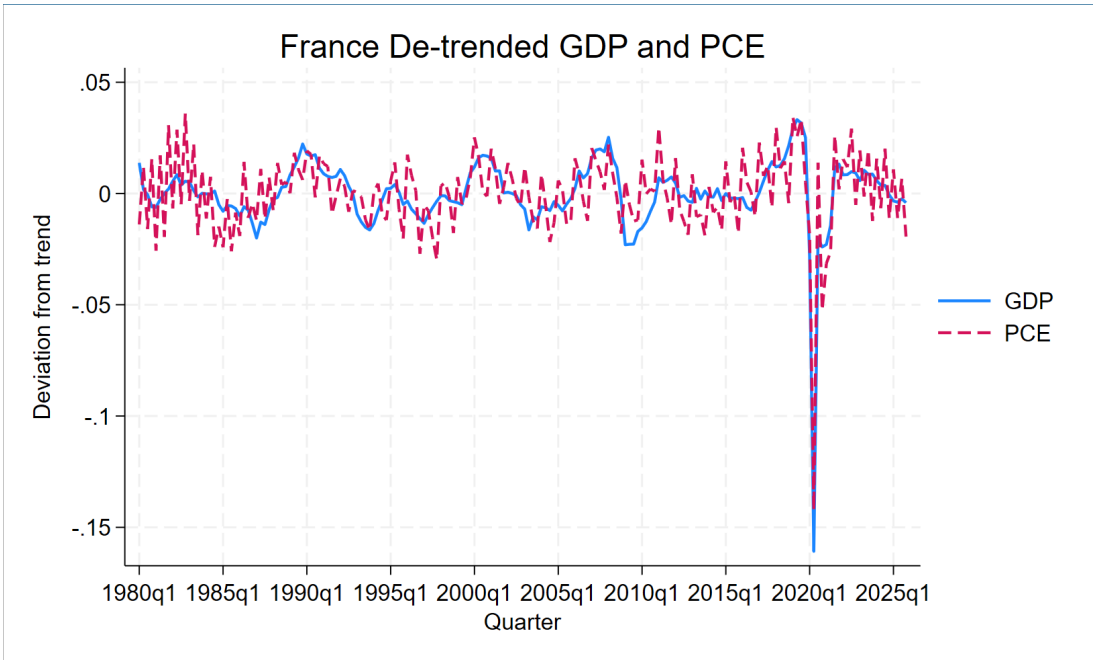

Figure 2: De-trended GDP and consumption in France (HP filter, $\lambda = 1600$).

Table 3: Covariance matrix (France)

	fr_pce_cycle	fr_gdp_cycle
fr_pce_cycle	0.000333	
fr_gdp_cycle	0.000208	0.000263

Table 4: Correlation matrix (France)

	fr_pce_cycle	fr_gdp_cycle
fr_pce_cycle	1.0000	
fr_gdp_cycle	0.7021	1.0000

Comment. Figure 2 suggests a weaker visual co-movement between the de-trended series for GDP and consumption in France than in the U.S. case. This is partly due to the fact that the consumption series initially exhibited seasonal fluctuations, and even after adjustment it still appears more irregular. Nevertheless, Table 4 reports a positive

correlation of 0.7021, indicating that the two cyclical components still move together to a substantial extent. As in the U.S. case, apart from the large drop during the Covid-19 period, both series appear relatively stable, which is coherent with the small values reported in the variance-covariance matrix in Table 3.

Table 5: Variance (diagonal) and correlation (lower triangle) – France and United States

	FR PCE	FR GDP	US GDP
FR PCE	0.000333		
FR GDP	0.7021	0.000263	
US GDP	0.3145	0.6107	0.000189

The combined variance-correlation table reveals two main findings. First, regarding volatility, French PCE displays the highest variance (0.000333), exceeding both French GDP (0.000263) and U.S. GDP (0.000189), which partly reflects the residual irregularity in the French consumption series even after seasonal adjustment. Second, regarding co-movement, the correlation between French PCE and French GDP (0.7021) is strong, though weaker than the U.S. counterpart reported in Table 2 (0.8718), consistent with the visual evidence in Figure 2. More interestingly, the cross-country correlations show that French GDP comoves more strongly with U.S. GDP (0.6107) than French PCE does (0.3145). This asymmetry is economically intuitive: output is more exposed to international trade and financial linkages, which tend to synchronize business cycles across countries, while consumption is more tied to domestic factors such as household income, credit conditions, and national fiscal policy. Overall, the results are broadly consistent with standard international business cycle facts — positive but imperfect cross-country correlations, with output more synchronized than consumption across borders.

Note: Seasonal Adjustment (France PCE). The French personal consumption expenditure series exhibited notable seasonal fluctuations, due to the fact that the raw FRED series is not seasonally adjusted. To address this, I applied a simple deterministic seasonal adjustment prior to HP filtering. Specifically, I regressed the log of French PCE on a full set of quarterly dummies:

$$\ln C_t^{FR} = \alpha + \sum_{q=2}^{4} \delta_q \cdot \mathbf{1}[quarter_t = q] + \varepsilon_t$$

and used the residuals $\hat{\varepsilon}_t$ as the seasonally adjusted series. This procedure removes the mean seasonal pattern from the log-level series by demeaning each quarter separately. The HP filter with $\lambda = 1600$ was then applied to these residuals to extract the cyclical component. Note that since OLS residuals are mean-zero by construction, the adjusted series is centered around zero before filtering, which does not affect the cyclical component but should may be kept in mind when interpreting levels.

Exercise 2 – Local Projection and Macroeconomic Shocks

Replication of Ramey (2016)

The shock I considered is the Ramey military news shock, which measures announcements of future government spending related to military events. It is normalized by lagged trend

GDP and the GDP deflator:

$$shock_t = 100 * \frac{rameynews_t}{y_{t-1}^{trend} * pgdp_{t-1}}$$

So it is not just standardized — it is scaled by the size of the economy.
The dependent variable is:

$$y_t = \frac{realGDP_t}{trendGDP_t}$$

where: real GDP $= 100 * \frac{ngdp}{pgdp}$, trend GDP is estimated using a quadratic trend in log real GDP. Thus, the outcome is real GDP normalized by trend GDP. Moreover, the sample period is: 1947Q1 – 2015Q4 and the controls included are the following: lags of the shock, lags of output, lags of government spending, lags of taxes, a time trend t and a quadratic trend t^2.

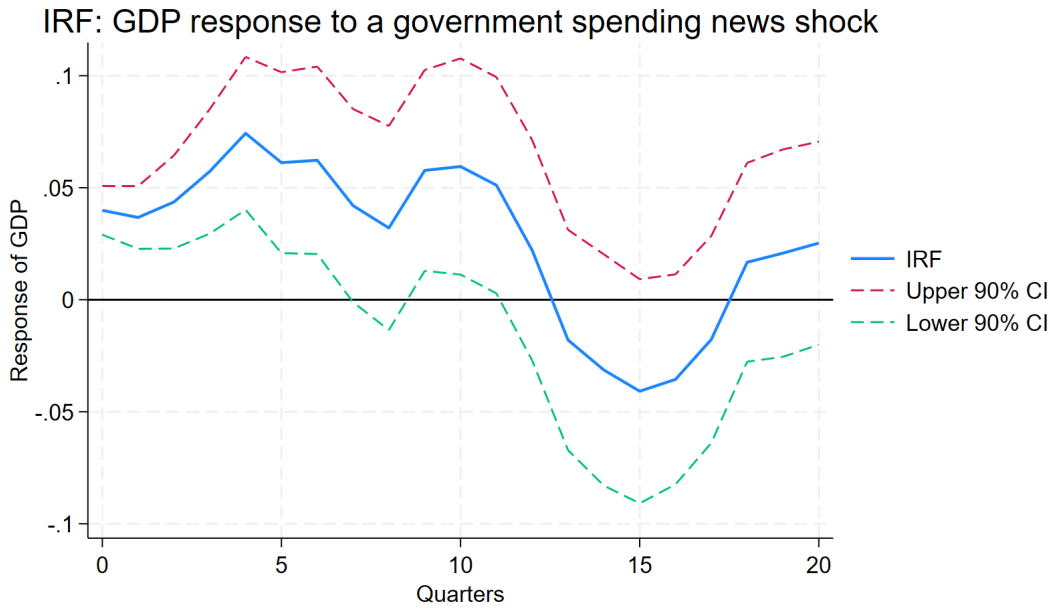

Figure 3: IRF of spending news shock on gdp.

Comment. Figure 3 reports the impulse response function of real GDP to the Ramey military news shock, estimated via local projections over the sample 1947Q1–2015Q4. The IRF displays a hump-shaped positive response: GDP rises gradually following the shock, reaching its peak response of approximately 0.05–0.06 around quarters 5–7, before declining and eventually turning non-significant. The initial positive response is statistically significant at the 90% level for several quarters around the peak, while the negative portion at longer horizons is not statistically distinguishable from zero. The qualitative pattern is broadly consistent with Ramey (2016), though some differences in the magnitude of the medium-run response may reflect minor differences in data vintage or lag specification. Ramey (2016): a delayed but persistent positive GDP response to an anticipated increase in government spending, followed by a reversion. The hump shape reflects the news nature of the shock — since the military build-up is announced before it materializes, GDP begins rising as spending is gradually implemented. Overall, the replication

is broadly successful, with the point estimates and confidence bands closely tracking those reported in the original paper.

Ramey (2016) and Romer & Romer tax shocks (2010)

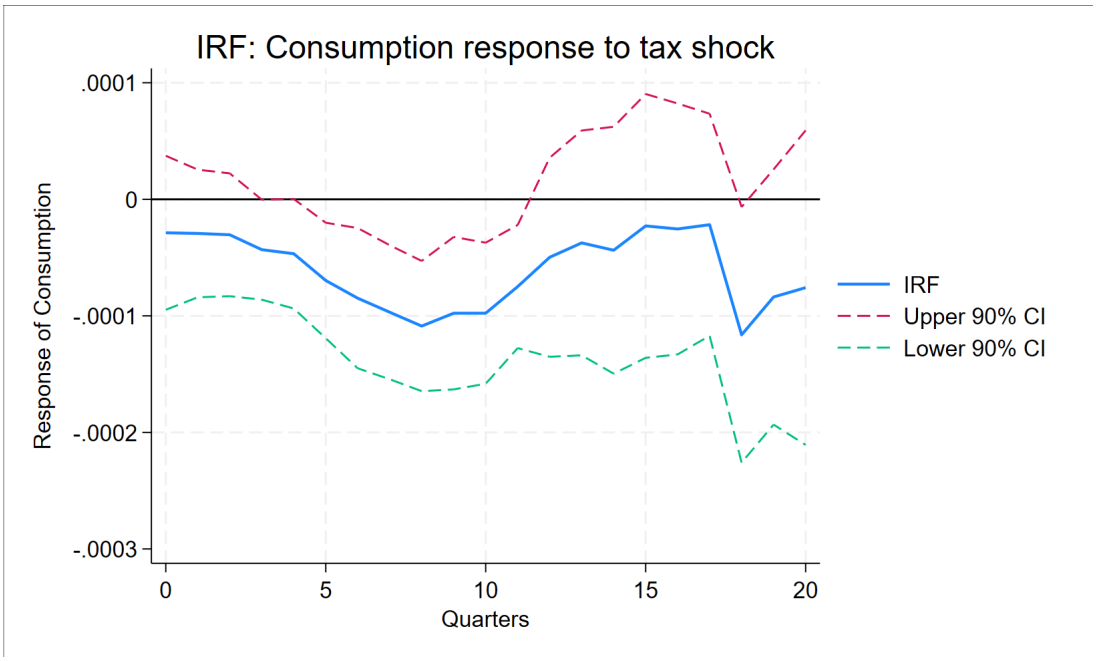

Figure 4: Impulse response function of consumption to a tax shock

Description. The figure reports the impulse response function (IRF) of consumption to an exogenous tax shock. The estimates are obtained using local projections following the empirical framework in Ramey (2016), combined with the narrative tax shock series constructed by Romer and Romer.

The estimated specification is:

$$c_{t+h} = \alpha + \beta_h \, taxshock_t + \gamma X_t + \varepsilon_{t+h}$$

where c_{t+h} denotes consumption at horizon h, $taxshock_t$ is the exogenous tax shock identified by Romer and Romer, and X_t represents the set of control variables. The horizon considered is $H = 20$ quarters. The control variables include lagged values of the tax shock, output, government spending, and taxes, as well as a linear and quadratic time trend. These controls follow the specification commonly used in the fiscal policy literature in order to account for macroeconomic dynamics and slow-moving trends.

The tax shock is taken from the narrative dataset of Romer and Romer, which identifies exogenous changes in tax policy based on historical policy records. In particular, the series measures legislated tax changes that are not systematically related to contemporaneous macroeconomic conditions, making them suitable for identifying causal effects of taxation.

The confidence intervals shown in the figure correspond to the 90 percent level.

Interpretation. The IRF suggests that an increase in the tax shock leads to a decline in consumption. In the very short run the effect appears small, while in the medium run consumption decreases more noticeably. This pattern is consistent with standard economic intuition: an increase in taxes reduces disposable income and therefore lowers consumption, holding constant the marginal propensity to consume.

At longer horizons the effect gradually diminishes and becomes statistically indistinguishable from zero, suggesting that the impact of the tax shock on consumption is temporary.

Overall, the results indicate that an increase in taxes tends to reduce consumption, although the estimated effects are relatively small and the confidence bands include zero for several horizons. The qualitative pattern of the response is nevertheless consistent with the theoretical prediction that higher taxes reduce household disposable income and therefore dampen consumption.

In conclusion, the empirical evidence is broadly aligned with the expected effect of tax shocks on consumption. The main difference with prior expectations is that the short-run impact appears relatively small, while the largest negative response occurs with some delay in the medium run.